SYNECHRON L2 — PLATFORM ENGINEERING CHEAT SHEET

PLATFORM = GOLDEN PATHS + SELF-SERVICE + AUTOMATION + SECURITY + OBSERVABILITY + RELIABILITY + GOVERNANCE

1. WHAT IS PLATFORM ENGINEERING?

Build and operate reusable internal capabilities so application teams can deliver safely, quickly and consistently. Think **golden paths, self-service, automation, security, observability and reliability**.

2. PLATFORM vs DEVOPS vs SRE

DevOps: collaboration + automation across development and operations. **Platform Engineering:** productises reusable internal capabilities and golden paths. **SRE:** engineering applied to operations, measured through SLOs, error budgets, observability and incident management.

3. GOLDEN PATH

A recommended supported route for a common task. Example: **service request** → **repo** → **CI/CD** → **container** → **Kubernetes** → **secrets** → **monitoring** → **dashboard/alerts**. Make the secure supported path the easiest path.

4. SELF-SERVICE PLATFORM

Developer → **approved template** → **Terraform/module** → **CI/CD** → **environment** → **deployment** → **observability**. Add RBAC, guardrails, auditability and standardisation.

5. PLATFORM CAPABILITIES

Git/PR: protected branches • **CI/CD:** Jenkins, GitHub Actions, Argo CD • **IaC:** Terraform, Ansible • **Runtime:** Kubernetes, Docker, EKS/AKS • **Observability:** Prometheus, Grafana, Splunk, CloudWatch/Azure Monitor • **Security:** IAM/RBAC, secrets, SAST/SCA, image scanning, NetworkPolicies • **Operations:** ServiceNow, runbooks, incident/change/problem management.

6. PLATFORM GOLDEN-PATH FLOW

Code → **PR** → **CI/security** → **immutable artifact** → **registry** → **IaC/environment** → **deployment** → **smoke test** → **observability** → **feedback**

7. RELIABILITY / OPERATIONS

A platform must be **deployable AND supportable**: SLI/SLO, error budgets, Golden Signals, actionable alerts, runbooks, capacity, recovery, incident response and RCA. Automate repetitive toil.

8. KUBERNETES PLATFORM CHECKLIST

Capacity → node groups → VPC/CNI → Services/Ingress → NetworkPolicies → RBAC → Secrets → requests/limits → probes → PDB → autoscaling → logs/metrics → upgrades → DR.

9. SECURITY & GOVERNANCE

Least privilege, workload identity, protected branches, secret management, dependency/image scanning, audit trails, approved IaC modules, change management and segregation of duties. Automation accelerates controls; it does not bypass mandatory controls.

10. TRICKY L2 QUESTIONS

“Did you build a complete IDP?” Be precise: “My experience is stronger in operating and improving shared cloud/Kubernetes platforms, standardised deployment, automation, observability and operational capabilities. I haven't owned a complete IDP product end-to-end.”
“Is Platform Engineering just DevOps?” “There is overlap, but platform engineering productises reusable capabilities and golden paths.”
“What would you automate first?” “High-volume, repetitive, standardisable work: service templates, CI/CD, IaC, environment configuration, secrets integration, observability and deployment validation.”

11. 45-SECOND SPOKEN ANSWER

“My understanding of Platform Engineering is that the platform team provides reusable, secure and observable capabilities that application teams consume through standard golden paths. In my experience, that includes cloud and Kubernetes platforms, Terraform-based infrastructure, CI/CD, deployment automation, access controls, monitoring and operational runbooks. I focus on standardisation, self-service where practical, guardrails and reducing operational toil. The platform should not only make deployment easier; it should also make services secure, observable, reliable and supportable in production.”